

Response to the reviewers

Manuscript: *Minimum Weighted Hazard-Exposure Dispatch: A Scheduling View with Verified Algorithms and a Wildfire Case Study* (formerly *Minimum Weighted Hazard-Exposure Dispatch: Complexity, an Exact Algorithm, and an FPTAS*)
Journal of Combinatorial Optimization — major revision

Dear Editor and Reviewers,

Thank you for the careful reading and the constructive reports. We revised the manuscript substantially. Reviewer 1’s central objection, that the theoretical results follow from the classical theory of $1\|\sum w_j U_j$, is correct, and we have rewritten the title, abstract, introduction, related work and summary table to say so plainly, to cite the classical source for each result, and to separate what is classical from the few things we believe are new. Reviewer 1’s question about the FPTAS exposed a real weakness in our experiments (the FPTAS’s rounding was inactive in most of them), which we explain and fix. Reviewer 2’s requests for a clearer hardness proof, a self-contained proof of the equal-cost theorem, a shorter positioning section and a discussion of parameter uncertainty are all addressed.

While revising we re-checked every numerical result, every cited source and the proofs (the statements and proofs are also machine-checked in Lean 4, with the scope stated in the Data Availability section), and we found several errors of our own in the version you reviewed; we list them in Section 3. The most consequential are: the case study used hazard times that were partly extrapolated although the cited NIST report documents them, and rested on a modelling premise that its own geography contradicts; and the earlier discussion of several vehicles contained an incorrect inference. All tables and figures were regenerated.

Numbering. Items are numbered in order of appearance, so the numbers of the version you reviewed have shifted. That version $\rightarrow$ now: Lemma 1 $\rightarrow$ Lemma 2; Theorem 2 (hardness) $\rightarrow$ Theorem 3; Theorem 3 (dynamic program) $\rightarrow$ Theorem 5; Theorem 4 (FPTAS) $\rightarrow$ Theorem 7; Theorem 5 (equal cost) $\rightarrow$ Theorem 8; Proposition 6 (naive deadline order) $\rightarrow$ Proposition 9. New in the revision: Proposition 1 (the round-trip model is conservative), Fact 1, Proposition 4 (constant hazard-arrival time), Lemma 6 (value-indexed table), Propositions 10 and 11 (greedy repair; FPTAS tightness), Theorem 12 (classification) and, in the new Section 4.7 (hazard geometry) and Appendix C, Theorems 13–24. Below, “Theorem a (b)” means Theorem a now and Theorem b in the version you reviewed. Section numbers refer to the revised manuscript; the Camp Fire study is Section 5.7.

Summary of the main changes

1. **Positioning and novelty** (title, abstract, Sections 1, 2, Table 2, conclusion): every result that is classical is labelled as such with its source (Karp 1972; Lawler and Moore 1969; Sahni 1976; Gens and Levner 1981; Lawler 1976; Dessouky et al. 1990; Moore 1968), and the claims to novelty are cut down to what we can support.
2. **What we now claim** (Section 4 and Table 2): a refined, exactly tight analysis of the FPTAS (Theorem 7, Proposition 11); hardness under the arrival reading even when the hazard-arrival time is constant (Proposition 4); the observation that the depot round-trip model is conservative

with respect to a chained route (Proposition 1); and elementary worst-case examples for natural heuristics (Propositions 9 and 10). The equal-cost theorem (Theorem 8) now covers vehicles with different speeds and is labelled classical; the classification (Theorem 12) is labelled an elementary corollary.

3. **Hazard geometry** (new Section 4.7, Appendix C): in answer to the requests for a result that is not inherited from scheduling theory we quantify what the round-trip abstraction costs. The ratio of the best chained route to the best round-trip plan is at most n and the bound is attained, by clustered sites and by a front that chases the crew along a road (Theorem 13); a speed-augmentation bracket in terms of a clustering ratio, which cannot be improved in general (Theorem 14), and a two-parameter bound (Proposition 15); a chained route on a line is pseudo-polynomial, has an FPTAS (Theorem 17; a selective version of a dynamic program of Tsitsiklis 1992) and is NP-hard with weights (Theorem 18); and for deadlines induced by a front at constant speed we classify the complexity by the slope and offset of the deadline as a function of the dispatch time (Proposition 19, Theorems 20–21), which shows that even a front that starts at the depot gives an NP-hard problem. Finally, for a front slower than the crew we give a polynomial-time approximation algorithm for the chained route whose ratio depends only on the front’s speed (Theorem 22; it uses a point-to-point orienteering algorithm from the literature as a subroutine and says nothing about general deadlines, for which only an $O(\log n)$ guarantee is known), show that the round-trip surrogate cannot give such a guarantee at any fixed speed (Proposition 23), and prove that the unit-weight chained problem is strongly NP-hard even for equidistant sites (Theorem 24), so no scheme better than a constant is possible. These statements are checked against brute force (`verify_geometry.py`); they are *not* formalised in Lean, and the manuscript says so.
4. **Hardness proof** (Theorem 3 (2)): a direct reduction from PARTITION with an explicit lemma (Fact 1) on feasibility under a common deadline.
5. **Experiments** (Section 5): redesigned, with exhaustive-search verification, instance sizes to $n = 400$, a stronger baseline, FPTAS experiments in which rounding is active, an adversarial family, and a comparison with a general-purpose MIP solver.
6. **Case study** (Section 5.7): rebuilt on hazard-arrival times documented by NIST for every community (nothing is extrapolated), with a comparison against a chained-route model and a sensitivity analysis with a stated protocol; the conclusions drawn are now conditional and modest.
7. **Discussion** (Section 6): corrected multi-vehicle subsection; parameter uncertainty in a new subsection with the two suggested references.
8. **Reproducibility**: all code and a Lean 4 formalisation of the theory are public (Data Availability).

1 Reviewer 1

While the manuscript is technically correct and generally well presented, the level of novelty of the theoretical contributions is not sufficiently convincing. [...] most of the main theoretical results appear to follow from well-known results in the scheduling literature. [...] Consequently, the manuscript does not clearly establish new complexity results, new algorithmic techniques, or

substantially new structural properties beyond those already known for the equivalent scheduling formulation.

Response. We agree. The manuscript proves that MWHED is $1 \parallel \sum w_j U_j$ and the general results follow from it. We have not tried to argue otherwise. Concretely:

- The title, abstract and Section 1 now say that the basic complexity picture and the algorithms are inherited from scheduling theory and that we claim no new technique for the general problem. The contributions are split into a list of classical results restated with complete proofs (hardness, the dynamic program, the FPTAS, the equal-size greedy, the classification) and a list of modest additions.
- Section 2 states, for each result, which classical result it is: Theorem 5 (3) is the Lawler–Moore dynamic program; Theorem 7 (4) is the classical value-scaling scheme for job sequencing with deadlines (Sahni 1976, Gens and Levner 1981); the hardness is Karp’s (Karp, 1972) and the knapsack equivalence; and Theorem 8 (5) is the scheduling of unit-time jobs on identical or uniform machines, for which $O(n \log n)$ algorithms are known (Lawler, 1976; Dessouky et al., 1990). The extension of Theorem 8 to several vehicles, with different speeds, is likewise classical, and we say so.
- The summary table (Table 2) has a column giving the origin of each result (classical, Lawler–Moore, Moore, elementary corollary, new).
- We removed the claim that the dynamic program and the FPTAS are “necessary” for any guarantee: Propositions 9 and 10 show only that deadline-order dispatch and one greedy repair have no guarantee.

1. Could the authors identify a new structural property?

Response. Yes, but they are modest, and we present them as such.

- **Proposition 4 (constant hazard-arrival time).** (To our knowledge not stated before; it is elementary.) Under the arrival reading of Remark 1 the natural homogeneous case is a hazard that reaches all sites at the same time, which is *not* a constant deadline d_i (it is $d_i = H + p_i/2$). We prove that MWHED stays weakly NP-hard there, even with all p_i even and $w_i = p_i/2$; constant p or constant w remain polynomial, so the restriction cannot be sharpened.
- **Proposition 1 (the round-trip model is conservative).** If the sites lie in a metric space, a crew that chains the sites reaches each site no later than in the depot round-trip model, so the round-trip optimum is a lower bound on what a crew can protect. This makes explicit when the premise of the model is costly (clustered sites) and is quantified on our case study.
- **Hazard geometry (Section 4.7).** The most specific new material: the round-trip surrogate can lose a factor n against a chained route (attained), with a bracket controlled by a clustering ratio (Theorems 13 and 14); chained routing on a line is pseudo-polynomial and has an FPTAS but is NP-hard with weights (Theorems 17 and 18); a constant-factor approximation for a front slower than the crew, with strong NP-hardness of the unit-weight problem (Theorems 22 and 24); and for affine deadlines $d = \beta + \alpha p$ (a constant-speed front) the complexity is classified, with

NP-hardness for $\alpha > 1$ even at $\beta = 0$, i.e. for a front that starts at the depot (Theorems 20 and 21). The proofs use exchange arguments and Partition reductions and are not deep; we claim them as new only in the sense that we found no source stating them (the line algorithm itself is a selective version of Tsitsiklis’s dynamic program, and we say so).

- **A sharper, exactly tight FPTAS analysis** (Theorem 7, Proposition 11): the scheme of Theorem 7 guarantees a constant $\rho_{n,\epsilon}$ that exceeds both $1 - \epsilon$ and $1/(1 + \epsilon)$, and the constant is attained (two extremal families), so the classical $1 - \epsilon$ bound is not attained.
- **The classification** (Theorem 12) remains, but we now call it an elementary corollary, and we note that the natural tractability boundary in the scheduling literature is Lawler’s agreeable-weights case (Lawler, 1976), which contains both constant p and constant w .

2. The manuscript repeatedly states the dynamic program and FPTAS as major contributions. However, the relationship between these algorithms and classical Lawler-Moore-type methods is not discussed in sufficient detail. Are the algorithmic contributions truly different from known scheduling algorithms?

Response. No. Theorem 5’s dynamic program is the Lawler–Moore dynamic program (jobs in due-date order, state equal to the accepted processing time, one test per accepted job); Remark 2 says so and describes exactly what our proof adds (a derivation from the earliest-deadline-first lemma in the notation of the dispatch problem). Theorem 7 is the classical value-scaling scheme of Sahni (1976) and Gens and Levner (1981); the manuscript states this, notes that faster schemes are known, and keeps the classical construction because its analysis is the cleanest and, with Theorem 7 and Proposition 11, exactly tight. Neither algorithm is presented as a contribution; they are a self-contained treatment for a transportation audience.

3. Why does FPTAS = 1000 always appears in every table? Could the authors strengthen the empirical evaluation?

Response. This was a genuine weakness. With weights at most 100 the scaling factor $K = \max(1, \epsilon w_{\max}/n)$ equals 1 in almost all of our runs, so no rounding took place and the “FPTAS” coincided with the exact value-indexed dynamic program. A ratio of 1.000 was therefore guaranteed and said nothing about the scheme. We state this explicitly (Section 5.2, last row of Table 4: the share of instances with $K = 1$; here n is the number of sites after deleting individually infeasible ones) and have redesigned the study (every table was regenerated):

- **Verification** (Section 5.1): every algorithm is checked against exhaustive search, including all $n!$ dispatch orders for $n \leq 7$ (400 instances), best-feasible-subset enumeration for $n \leq 14$ (1,500 instances, one third with weights up to 10^6 , FPTAS guarantee checked in 4,500 runs of which 2,923 had active scaling), and exhaustive search over assignments for the equal-cost algorithm with $m = 2, 3$ vehicles; the added results are checked by a further script (1,600 instances for the different-speed theorem, 600 for Proposition 4, 4,400 random instances plus an adversary for the refined FPTAS bound, and recursions cell by cell). No violations.
- **Accuracy across sizes** (Section 5.2): n up to 400, 50 instances per size, worst-case ratios as well as means, and a stronger baseline, EDD with skipping, in addition to naive EDD.

- **FPTAS with active scaling** (Section 5.3): weights up to 10^6 (K in the hundreds to thousands), a strongly correlated family, and an adversarial family. On random and strongly correlated instances the FPTAS is nearly exact (worst ratio 0.9993 at $\epsilon = 0.5$) because the bound is pessimistic when the optimum contains many sites; on the adversarial family the ratio falls to 0.669 at $\epsilon = 0.5$ and follows the guarantee. We now state that the adversarial column is a single deterministic instance equal to a closed form, not an empirical worst case. A simple completion step repairs the family completely, and the sharpened bound of Theorem 7 concerns the scheme without that step.
- **Runtime** (Section 5.4) re-run with active scaling, and a **comparison with a general-purpose solver** (Section 5.5): HiGHS on the subset formulation returns the same optimum on all 40 instances and is much slower than the dynamic program (between about 15 and 190 times on these instances; the solver’s timings are noisy).

2 Reviewer 2

3. *The part of MWHEd’s positioning in page 6 is comprehensible, nevertheless it could be rephrased better by shorter arguments.*

Response. The “Positioning” paragraph (Section 2) was shortened and now lists the literatures it draws on and refers to Table 2; the two long paragraphs on real-time schedulability were cut to a short one, and the sentences on recent matroid-partition and multiagent-matroid papers were removed, since they carried no result used later.

1. *Theorem 2 on NP-hardness of MWHEd might require a bit efforts to get full clarity. In the 3rd line from the bottom of page 9, don’t you think that the fact “completion times along a fixed set are additive” is too strong? What fixed sets should we use here? [...] The authors should explain about additivity of special sets [...] since it is not obvious by default, feasibility condition does not explicitly implies additivity. [...] the above condition ($\diamond$) $\sum_{i \in S} p_i \leq D$ is crucial to relate your formulation to a 0/1 knapsack instance.*

Response. Thank you; the sentence was imprecise. We replaced it by an explicit statement, **Fact 1** (Section 4.1), with a proof: *if all deadlines equal D , a set S of sites can all be served on time iff $\sum_{i \in S} p_i \leq D$* , which is exactly your condition ($\diamond$). The “fixed set” is an arbitrary subset S . The only additivity used is that the *last* site served from S completes at the sum of all of S ’s occupancies, because the vehicle is never idle and each p_i does not depend on the predecessor; this gives necessity. For sufficiency every completion time is a partial sum of positive numbers and so is at most the total. The proof of Theorem 3 cites Fact 1, and the knapsack correspondence for general weights is stated after the theorem. We also made the reduction handle odd total A and sites with $a_i > A/2$ explicitly.

2. *[...] your claim on “NP-complete of the 0/1 knapsack problem” comes from polynomial reduction of PARTITION (one of the six popular graph-based NP-complete problems [...]). This is a solidly smart argument, if the above concern is clearly elucidated. Otherwise, a graph-based pure proof for Theorem 2 possibly might give better insights (ref. Delazeri and Ritt (2026)).*

Response. We removed the dependence on the knapsack citation: Theorem 3 gives a *direct* polynomial reduction from PARTITION to MWHED ($p_i = w_i = a_i$, $d_i = A/2$), so that the proof is self-contained and shows hardness when $w_i = p_i$; membership in NP is added. We did not give a graph-based proof: MWHED has no graph structure once each visit is a round trip from the depot, so there is no natural graph on which to base one. We do note (Remark 4 after Theorem 3) that the hard instances are realized by a tiny network, $K_{2,n}$ with a hazard front that reaches all sites simultaneously. Delazeri and Ritt (2026) prove NP-completeness for a different, graph-based model of wildfire suppression with simultaneous resources; Section 2 says so.

3. [...] the authors would elucidate more explicitly Theorem 5’s proof. In particular, would you remind readers the concepts (a) transversal matroid, union-find structure, and (b) the related notation $\alpha(n)$?

Response. Section 4.4 opens with a paragraph recalling the notions used: matroid and its three axioms, the greedy algorithm, the transversal matroid (sites against (vehicle, position) slots), and the union–find structure, with α defined as the inverse Ackermann function. The proof of Theorem 8 (5) is in five steps: (1) slots; (2) a counting criterion for feasibility; (3) a direct proof of the matroid augmentation property; (4) a short proof that greedy is optimal on any matroid; (5) the implementation, with a proof that it accepts a site exactly when the kept set plus that site is feasible. The theorem now allows vehicles with different dispatch times and is stated as classical. The union–find structure is described with union by size and path compression (which together give the $O(n\alpha(n))$ bound; path compression alone gives $O(\log n)$ amortized), including the label array needed for our linking direction. Algorithm 3 and Example 5 are rewritten accordingly.

4. [...] do we agree that the problem of MWHED is obviously data-driven, the proofs and optimal solutions obtained depend on both accurate formulation and precise calibration of model parameters, upon uncertainty? To some extent, would formulation of uncertainty [...] (at not just variable level but also at parameter one) be considered in next studies, where stochastic programming might be a potential approach? Kindly, e.g. refer to problems of evacuation planning under unsafe contexts and facility location under disruptive events (FLDE), ref. [1, 2].

Response. We agree. Section 6.3 (“Uncertain parameters”) covers uncertainty in p_i , d_i and w_i , not only in the deadlines. It discusses (i) scenario-based and two-stage stochastic formulations and which of our results survive (the earliest-deadline lemma holds scenario by scenario but not simultaneously; a worked example shows that sorting by expected deadline can be nearly twice as bad as optimal), citing your two references (Wang, 2020; Koca, 2023); (ii) a robust-feasibility observation; and (iii) calibration and sensitivity analysis. The case study (Section 5.7) now includes a Monte-Carlo sensitivity analysis over response speed, documented event times and departure delay, in an independent and in a correlated variant, with standard errors. The finding is conditional: the exact optimum inherits the assumed nominal speed. The plan optimal at 80 km/h (an optimistic corner) is fully on time in only 16.2% of 20,000 draws and retains 31.9% of the drawn optimum on average, whereas the plans optimal at any nominal speed below about 68 km/h, which coincide with a plan chosen against a sample of perturbed scenarios, are fully on time in 79.3% and retain 99.7%. The paper states that this is a property of the depot-spoke model. Stochastic programming for MWHED itself is listed as the main direction for future work; we have not attempted a solution method for it here.

3 Errors of our own corrected in this revision

In the course of revising we re-checked the manuscript, the code, the cited sources and the proofs, and found the following errors in the version you reviewed, which we correct and disclose.

1. **Case study: hazard times.** The fire’s timeline was stated as “52 and 71 minutes after ignition”, but the NIST report uses clock times and puts ignition at about 06:20, so the documented events (07:25 in Concow, 07:44 in Paradise) are 65 and 84 minutes after ignition; the figures we used are reproducible only with a start time taken from another source. Two of the four hazard times were *extrapolated* by an isotropic spread-rate rule (Magalia at 58 and Yankee Hill at 64 minutes), although the report documents events for them: spot fires in Old Magalia at 08:40 and the fire running through Yankee Hill on the following day; the extrapolation was far off. The ignition-point detail (transmission tower) is not in the NIST report. The case study now uses the documented clock times of the first events in the four communities, with time zero equal to the initial dispatch time in NIST’s dispatch log, states the events as the report words them, and uses NIST’s origin point in the figure. All case-study results were recomputed and the earlier numbers are superseded.
2. **Case study: modelling premise.** Concow, Paradise and Yankee Hill are within about 10 km of each other but 22–37 km from the depot, so the round-trip model was far from the geometry. The paper now proves that the round-trip model is conservative when travel times form a metric (Proposition 1), compares it with a chained route on the same data (Table 12), and says that the case study illustrates the model and does not support operational recommendations. We also state the protocol of the sensitivity analysis, add a correlated-error variant, report that the nominal plan depends on the assumed speed (it includes Concow only from about 68 km/h), and add a paragraph on the ethics of conceding low-weight sites and on what “protected” means.
3. **Case study: deadline semantics.** The model’s on-time test counts the completion of the round trip (return reading), but the narrative speaks of reaching the site (arrival reading). Remark 1 makes the two readings precise (the arrival reading is an instance of MWHEd with deadlines $d_i^{\text{haz}} + p_i/2$) and the case study uses it; the return reading is reported for comparison. Individually infeasible sites are deleted before every algorithm, including the naive baseline; statements that the optimum “saves” a number of people were replaced by the on-time weight.
4. **Naive baseline.** Our naive EDD baseline kept dispatching to sites that were already late, which made it a weak anchor; we added EDD with skipping.
5. **Several vehicles and release dates.** The claim that the $m = 1$ instance “embeds” into an instance with more vehicles by adding idle vehicles was wrong (extra vehicles can serve sites); we now use blocker sites. The inference from Heeger and Molter (2025) that the equal-cost case becomes hard with several vehicles was also wrong: that result concerns the unweighted problem on several machines *with release dates*, which MWHEd does not have, and for one machine equal processing times with release dates and weights are polynomial (Baptiste, 1999). Theorem 8 now shows tractability for every number of vehicles, and the text states exactly what the cited result covers.
6. **Heeger–Hermelin.** The text said that a small number of distinct dispatch costs “does not guarantee tractability”. The result is $W[1]$ -hardness in that parameter (no $f(k)\text{poly}(n)$ algorithm); the text now says this and makes no claim about two or more distinct values.

7. **Proofs.** In the FPTAS proof an inequality was used in the wrong direction in the main text (the appendix repaired it); the displayed recursions for the dynamic programs, written as an indicator times a value, were wrong as literally stated (they return 0 when the test fails) and are now written with cases; and the proof of the earliest-deadline-first lemma did not cover ties between deadlines. Each fix was checked against brute force (`verify_extensions.py`).
8. **Implementation.** The FPTAS guarantee relies on Assumption 1 (so that $w_{\max} \leq W^*$), which the code did not enforce; both algorithms now delete infeasible sites first, and the code was rewritten and checked against exhaustive search.
9. **Bibliography.** We verified every entry against a publisher, Crossref or arXiv record and corrected, among others: the author list and volume of Alvelos et al. (2025) (three authors, volume 86), the author order of the NIST report, and missing volume, article and page data for several proceedings entries.
10. Minor: the dash convention in the dynamic-program example table was described incorrectly; the explanation of why the FPTAS runtime was flat in ϵ cited saturating weights, whereas the cause was $K = 1$. Both are fixed.

We believe the revised manuscript is considerably more honest about what is and is not new, and we thank the reviewers for the comments that led to these changes.

Sincerely,
The authors

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